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Jayadev Joy

Education

2023–Present **Doctor of Philosophy**, *New York University*, Brooklyn, NY, United States, ***GPA: 4/4***
Major: **Electrical & Computer Engineering**, Advisor: **Dr. Sundeep Rangan**

2018–2023 **Dual Degree (B.Tech, M.Tech)**, *Indian Institute of Technology Madras*, TN, India, ***GPA: 9.13/10***
Major: **Electrical Engineering**, Advisor: **Dr. Srikrishna Bhashyam**

Internships

Summer 2026 **Digital Signal Processing Intern**, *Marvell Technology*, Santa Clara, CA, United States
- Built a scope-free TDECQ measurement flow, replacing DCA oscilloscope-based characterization.
- Worked on transmitter FIR optimization to improve link performance.

Summer 2021 **Software Engineering Intern**, *Barclays*, Pune, MH, India
- Collaborated with a team of 3 to develop a user-friendly web portal using Docker for file sharing.
- Implemented front-end development using HTML, CSS, and JavaScript.

Summer 2020 **Electrical Engineering Intern**, *Tan90 Thermal Solutions*, Chennai, TN, India
- Engineered a solar-powered cooling backpack using Arduino, for vaccine preservation.
- Designed a smart power distribution circuit, optimizing energy consumption & extending battery life by 15%.

Technical Skills

Programming Python (pandas, scikit-learn, PyTorch, TensorFlow, Keras, Sionna, Qiskit), C, MATLAB, Verilog.

Software Git, Jupyter, IBM Quantum, LaTeX, Linux, HPC, Simulink, Autodesk EAGLE, LabVIEW, LTspice.

Research Projects

Spring 2026 **Discrete-Time Modeling and Analysis of N-Path Receivers with Interference**, *NYU Wireless*
- Built an end-to-end discrete-time simulator of an N-path receiver, with amplifier saturation, and quantization.
- Derived closed-form I/O responses for signal and interference, and benchmarked against a conventional receiver.

Summer 2025 **Learning-Based Signal Recovery in Nonlinear RF Systems with Interference**, *NYU Wireless*
- Designed a learned MLVAMP algorithm with neural denoisers for nonlinear signal recovery under interference.
- Modeled receiver impairments and built a hybrid inference framework achieving 20 dB gain over linear methods.

Summer 2024 **Interpolation Techniques for Fast Channel Estimation in Ray Tracing**, *NYU Wireless*
- Developed an interpolation-based ray tracing to optimize channel modeling in Sionna.
- Implemented a reflection model for accurate LOS MIMO simulation, improving spherical wavefront modeling.

Fall 2022 **Sparse Signal Recovery for Massive Random Access**, *Communication Lab, IIT Madras*
- Improved a SSR algorithm, and integrated into a larger coding scheme (FASURA) for massive random access.
- Implemented SBL, MSBL, AMP-MSBL & GGAMP-MSBL algorithms from scratch.

Academic Projects

Spring 2024 **Music Classification Using Neural Networks**, *NYU Tandon*
- Built a neural network with Adam optimizer and early stopping, achieving 99% accuracy.
- Enhanced performance through feature extraction with Librosa and hyperparameter tuning.

Spring 2024 **KNN Voter Classification Using Exit Poll Data**, *NYU Tandon*
- Developed a custom KNN distance metric, improving voter classification accuracy with incomplete data.
- Enhanced model performance through feature selection, achieving over 84% test accuracy.

Fall 2023 **Markov Decision Processes with Controlled Observations**, *LARX, NYU Tandon*
- Analyzed a continuous-time MDP, integrating controlled actions and optimizing discrete-time observations.
- Extended the framework to a multi-agent system, exploring strategic considerations in a two-player context.

Fall 2022 **Maximum Likelihood Sequence Detector**, *Communication Lab, IIT Madras*
- Simulated a Viterbi algorithm sequence estimator, assuming knowledge of channel parameters.
- Evaluated the SER of transmitted symbols for a wide range of SNR values and decoding delays.

Fall 2021 **Parallel M/M/2 Queue**, *Communication Lab, IIT Madras*
- Simulated a two-queue system with customer arrivals following a Poisson process.
- Evaluated the error ($< 3\%$) and generated time plots of the number of customers in the queues.

Spring 2021 **Toxic Comment Classification**, *Communication Lab, IIT Madras*
- Evaluated the performance of naive Bayes, logistic regression, SVM & random forest classifiers.
- Performed text preprocessing & word embedding (TF-IDF) on the training dataset.

Publications

Oct 2026 **J. Joy**, R. Nikandish, *et al.*, "Discrete-Time Modeling and Analysis of N-Path Receivers: Interference and Nonlinear Distortion," *60th Asilomar Conference on Signals, Systems, and Computers*, CA, USA.

Oct 2024 R. Chen, **J. Joy***, *et al.*, "Interpolation Techniques for Fast Channel Estimation in Ray Tracing," *58th Asilomar Conference on Signals, Systems, and Computers*, CA, USA.

Honors & Achievements

2024 Awarded the **Ernst Weber Fellowship** in Electrical & Computer Engineering at NYU.

2023 Awarded the **School of Engineering Fellowship** in Electrical & Computer Engineering at NYU.

2023 Selected participant in the **IIT Bombay Computer Science Research Symposium**, 2023.

2019 Recipient of the **64th Foundation Scholarship** at IIT Madras during 2019-20 and 2020-21.

2018 Achieved **All India Rank 1518** (percentile of 99.9902) in IIT-JEE Advanced 2018.

2018 Achieved **All India Rank 1622** (percentile of 99.9985) in IIT-JEE Main 2018.

2018 Secured **Kerala State Merit Rank 48** (percentile of 99.9994) in KEAM Entrance Exam 2018.

2018 Accepted into the **B.Sc. (Hons.) Mathematics** program at Chennai Mathematical Institute, 2018.

2016 Received **National Talent Search Examination** (NTSE) Merit Certificate and Scholarship in 2016.

Relevant Coursework

EE/CS Machine Learning, Estimation & Detection Theory, Information Theory, Channel Coding, Queuing Theory, Wireless & Digital Communications, Quantum Computation, Multirate Digital Signal Processing.

MATH Probability Theory, Stochastic Modeling, Optimization, Linear Algebra, Statistics.

Additional Experience

2024–Present **Research Assistant**, *NYU Wireless*, Brooklyn, NY, United States
- Conduct research at NYU Wireless under the supervision of Dr. Sundeep Rangan.
- Focus on wireless communications, statistical signal processing, AI/ML in wireless, and information theory.

Fall 2024 **Course Assistant**, *ECE Department, New York University*, Brooklyn, NY, United States
- Assisted in LaTeX formatting and correction of tutorials for ECE2233 - Introduction to Probability.
- Supervised and graded MATLAB-based practical assignments and invigilated the final exam.

Spring 2023 **Teaching Assistant**, *NPTEL Office, IIT Madras*, Chennai, TN, India
- Conducted weekly interactive tutorial sessions for the graduate course EE5111 - Estimation Theory.
- Supervised quizzes and end-semester examinations.

2022–2023 **Teaching Assistant**, *Department of Electrical Engineering, IIT Madras*, Chennai, TN, India
- Led weekly tutorial sessions for EE5112 - Detection Theory, EE6133 - Multirate DSP, and EE2004 - DSP.
- Managed the invigilation and grading of assignments, quizzes, and exams.

Fall 2020 **Online Instructor**, *Camp K12*, Gurgaon, HR, India
- Taught app development using MIT App Inventor to kids aged 6-18 via live online sessions.
- Conducted live demo sessions on AI, app development, and 3D coding (AR, VR).

Spring 2019 **Academic Writer**, *Mathpresso Inc.*, Seoul, South Korea
- Prepared detailed solutions for 540 SAT mathematics questions over a duration of 3 weeks.
- Revised the solutions to meet client requirements and provide better comprehension.